FAMILY SIZE AND PARENTS' LABOR SUPPLY

Effect of Exogenous Variation in Family Size

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In short… what is the paper about?

- Use IV to estimate effect of childbearing on female labor supply in urban areas of Peru.
- Information from the National Household Survey (ENAHO) for 2004 – 2011.
- IV strategy based on sibling sex mix. Tries to exploit parental preferences for mixed sibling-sex composition.
- Interest on the effect of family size on participation on the formal job market.

What question(s) am I trying to answer?

- Assess the relationship between childbearing and labor supply.
- Given absence of a strong social security network in Peru, effects of family size on household wellbeing might be underestimated if larger family size negatively impacts probability of being on formal job market.

What data?

- Only urban households from the eight different regions identified in the ENAHO.
- Rural households are expected to have a larger size, independent of the sex of the initial siblings (IV strategy proposed would result in weak instruments).
- Geographical controls on state or region level (North coast, Central Coast, South Coast, North Highlands, Central Highlands, South Highlands, Jungle and Lima).

What is the methodology?

- Angrist and Evans (1998) instrumental variable (IV) strategy based on the sibling sex mix in families with two or more children.
- Estimation strategy involves use of 2SLS and OLS regression models relating labor-market outcomes to fertility and a variety of exogenous covariates (set of characteristics).

Fraction of Families that Had Another Child by Parity and Sex of Children

Sex of first child in families with one or more children	All women		Married women	
	Fraction of sample	Fraction that had another child	Fraction of sample	Fraction that had another child
(1) one girl	0.488	0.604	0.487	0.673
		(0.0030)		*(0.0032)*
(2) one boy	0.512	0.613	0.513	0.680
		(0.0029)		*(0.0031)*
difference (2) - (1)		0.009		0.007
		(0.002)		*(0.002)*

Sex of first two children in families with two of more children	All women		Married women	
	Fraction of sample	Fraction that had another child	Fraction of sample	Fraction that had another child
one boy, one girl	0.495	0.414	0.496	0.428
		(0.0038)		*(0.0042)*
two girls	0.243	0.453	0.244	0.472
		(0.0055)		*(0.0060)*
two boys	0.262	0.443	0.261	0.461
		(0.0053)		*(0.0058)*
(1) one boy, one girl	0.495	0.414	0.496	0.428
		(0.0038)		*(0.0042)*
(2) both same sex	0.505	0.448	0.504	0.467
		(0.0038)		*(0.0042)*
difference (2) - (1)		0.034		0.039
		(0.0000)		*(0.0000)*

Notes: Samples are the same as in Table 2. Standard errors are reported in parentheses.

OLS and 2SLS Labor Supply Models (I)

Estimation method	All Women					Married Women				
	OLS		2SLS			OLS		2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Instrument for More than 2 children	Same sex	Two girls	Same sex	Two boys, Two girls	Two girls	Same sex	Two girls	Same sex	Two boys, Two girls	Two girls
Dependent variable:										
Worked for pay	-0.0021	-0.0041	-0.0518	0.0100	0.1050	-0.0037	-0.0074	-0.0883	-0.0247	0.0275
	(0.0052)	*(0.0104)*	*(0.1302)*	*(0.1241)*	*(0.1399)*	*(0.0057)*	*(0.0114)*	*(0.1366)*	*(0.1247)*	*(0.1331)*
Formal job	-0.0068	-0.0137	-0.2170	-0.2550	-0.3560	-0.0123	-0.0247	-0.4090	-0.4500	-0.5470
	(0.0062)	*(0.0124)*	*(0.1974)*	*(0.1938)*	*(0.2324)*	*(0.0068)*	*(0.0135)*	*(0.2361)*	*(0.2325)*	*(0.2791)*
Hours/week	0.1580	0.3160	5.3050	-1.8050	-5.5670	0.0899	0.1800	3.0350	-5.9350	-8.4920
	(0.3135)	*(0.6270)*	*(10.6386)*	*(9.2618)*	*(9.6409)*	*(0.3463)*	*(0.6925)*	*(11.7297)*	*(9.7709)*	*(9.9859)*
Labor income	-7	-13	-171	817	2987	5	9	123	1366	2148
	(109)	*(217)*	*(2790)*	*(2715)*	*(3210)*	*(119)*	*(238)*	*(3067)*	*(2759)*	*(2918)*
ln(Family income)	-0.0050	-0.0101	-0.1260	-0.2440	-0.4260	-0.0024	-0.0048	-0.0571	-0.2380	-0.3870
	(0.0081)	*(0.0162)*	*(0.2018)*	*(0.1925)*	*(0.2180)*	*(0.0087)*	*(0.0174)*	*(0.2069)*	*(0.1892)*	*(0.2035)*

(Standard errors in parentheses).

First Stage Labor Supply Models (II)

OLS - First Stage Regression

Independent variable	Same Sex	2Boys,2Girls		Two Girls	Obs.	Same Sex	2Boys,2Girls		Two Girls	Obs.
Work	0.0399	0.0260	0.0538	0.0538	33,607	0.0418	0.0219	0.0617	0.0617	28,460
(s.e.)	*(0.0049)*	*(0.0069)*	*(0.0071)*	*(0.0071)*		*(0.0053)*	*(0.0075)*	*(0.0077)*	*(0.0077)*	
[R^2]	*[0.154]*	*[0.154]*		*[0.153]*		*[0.170]*	*[0.0170]*		*[0.170]*	
Formal job	0.0315	0.0234	0.0395	0.0395	19,878	0.0332	0.0216	0.0388	0.0388	16,184
(s.e.)	*(0.0065)*	*(0.0090)*	*(0.0093)*	*(0.0093)*		*(0.0071)*	*(0.0099)*	*(0.0103)*	*(0.0103)*	
[R^2]	*[0.146]*	*[0.146]*		*[0.145]*		*[0.146]*	*[0.164]*		*[0.164]*	
Hours week	0.0298	0.0129	0.0466	0.0466	23,240	0.0296	0.0091	0.0502	0.0502	19,117
(s.e.)	*(0.0060)*	*(0.0083)*	*(0.0086)*	*(0.0086)*		*(0.0066)*	*(0.0091)*	*(0.0094)*	*(0.0094)*	
[R^2]	*[0.150]*	*[0.150]*		*[0.150]*		*[0.168]*	*[0.168]*		*[0.168]*	
Labor Income	0.0389	0.0282	0.0495	0.0495	20,335	0.0388	0.0186	0.0590	0.0590	17,226
(s.e.)	*(0.0063)*	*(0.0089)*	*(0.0090)*	*(0.0090)*		*(0.0068)*	*(0.0096)*	*(0.0097)*	*(0.0097)*	
[R^2]	*[0.149]*	*[0.149]*		*[0.149]*		*[0.168]*	*[0.168]*		*[0.168]*	
Family Income	0.0399	0.0261	0.0538	0.0538	33,606	0.0418	0.0220	0.0617	0.0617	28,459
(s.e.)	*(0.0049)*	*(0.0069)*	*(0.0071)*	*(0.0071)*		*(0.0053)*	*(0.0075)*	*(0.0077)*	*(0.0077)*	
[R^2]	*[0.154]*	*[0.154]*		*[0.154]*		*[0.170]*	*[0.170]*		*[0.170]*	
With another covariates	Yes	Yes		Yes	- -	Yes	Yes		Yes	- -

Multinomial Labor Supply Models

Multinomial Regression
(Unemp=0 / Informal=1 / Formal=2)

multi_job	Coef.	Std. Err.
	0 (base outcome)	
1		
hhmorethan2	-0.0582	0.0276
hh1boy	0.0126	0.0252
hh2boy	-0.0548	0.0252
age	0.0924	0.0032
age_1birth	-0.0853	0.0043
year		
region		
_cons	-1.8178	0.0943

multi_job	Coef.	Std. Err.
2		
hhmorethan2	-0.5750	0.0381
hh1boy	0.0030	0.0338
hh2boy	-0.0093	0.0337
age	0.0976	0.0042
age_1birth	0.0127	0.0053
year		
region		
_cons	-5.9015	0.1429

Multinomial Labor Supply Models

Multinomial Regression
(Unemp=0 / Informal=1 / Formal=2)

multi_job	Coef.	Std. Err.
	0 (base outcome)	
1		
hhsamesex	-0.0004	0.0252
hh1boy	0.0128	0.0252
hh2boy	-0.0545	0.0252
age	0.0900	0.0030
age_1birth	-0.0824	0.0040
year		
region		
_cons	-1.8280	0.0948

multi_job	Coef.	Std. Err.
2		
hhsamesex	-0.0230	0.0336
hh1boy	0.0072	0.0336
hh2boy	-0.0026	0.0336
age	0.0764	0.0040
age_1birth	0.0407	0.0050
year		
region		
_cons	-6.0488	0.1436